

Financial Markets: 11. Behavioral Finance and the Role of Psychology

Robert J. Shiller

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Original lecture by Robert J. Shiller. Transcript-derived notes curated by [LazyingArt LLC](#).

Chapter 1

Behavioral Finance and the Role of Psychology

In this lecture Robert J. Shiller asks us to widen finance until it includes the actual people who inhabit it. Rationality remains useful; the lecture never denies that. But rationality is only a partial language for institutions whose success depends on minds, moods, habits, framing, reputation, and character. Behavioral finance therefore appears here not as a rejection of finance theory, but as a correction to an overnarrow picture of human conduct.

These notes follow Lecture 11 of the Yale Financial Markets series, curated by LazyingArt LLC. The lecture unfolds in a deliberate order. It begins with a moral and institutional preface, then slows down for the best-known formal core in behavioral economics, and only then accelerates into a wider catalog of recurring psychological mechanisms that matter for markets.

1.1 Behavioral Finance and the Limits of Rationality

Shiller begins by saying quite plainly that behavioral finance has become an important element of modern finance, even if it remains controversial. The controversy is itself revealing. Economists have long relied on rationality as a simplifying principle, and that principle has indeed been useful. But the lecturer's point is that its usefulness is limited. People are often rational, not completely rational. They can be intelligent in many dimensions and still make serious mistakes.

That point matters in finance because financial institutions are designed for real people, and their functioning depends on the behavior of real people. The lecture places this alongside a broader scientific development: a revolution in neuroscience. The human brain is not simple, so one should not expect the economy built out of human decisions to be simple either.

Already in the opening minutes the lecture introduces the darker implication. We are aware of one another's weaknesses, and that awareness can create an impulse to exploit. If we see somebody behaving foolishly, we may be tempted to turn that error to our own advantage. That is the danger that motivates the entire lecture. But before the lecturer catalogs any particular bias, he insists on placing that danger in perspective.

1.2 Morality, Exploitation, and Character

The lecture's first real pivot is moral. We are told to expect a discussion of human failings, but we are also told not to begin from cynicism. The business world, Shiller says, does not exploit human weakness as relentlessly as one might expect. Successful businesses have long-run interests. They care about reputation. A strategy that is blatantly exploitative may generate short-run gain, but it works against durable advantage.

He adds a second restraint: morality itself is part of human behavior. On the lecturer's telling, we do not merely maximize against one another. We also have an evolved impulse to be moral, and that impulse limits how much satisfaction we get from exploiting the mistakes of others. The lecture never says manipulation is absent. It says the market system contains more moral and institutional restraint than a shallow cynicism would predict.

1.2.1 Question & Answer

Question. If people are imperfect and manipulable, why are financial institutions not simply machines for exploiting them?

Answer. The lecture's answer is cumulative rather than doctrinaire. Exploitation is checked by long-run reputation, by repeat business, by trust, by regulation, and by the ordinary desire not to be the sort of person who preys on others' mistakes. Opportunities for exploitation are real, but they do not exhaust the structure of finance. This is why the lecture can honestly turn to human error without concluding that markets are nothing but organized predation.

To put that optimistic counterweight on firmer ground, the lecture turns to Adam Smith.

1.2.2 Adam Smith and Praiseworthiness

Shiller returns not only to *The Wealth of Nations* of 1776, but to the earlier *Theory of Moral Sentiments* of 1759. That move is essential to the lecture's rhythm. Before we see any value function or weighting function, we are asked how an economy can work if people are selfish, vain, and sometimes manipulative.

Smith's answer, as presented here, begins with praise. Human beings want approval. We crave the good opinion of others. But Smith's deeper observation is that mature people do not stop there. They move from wanting praise to wanting praiseworthiness. We do not merely want to be admired; we want to deserve admiration. That is the step that makes an economy workable.

The lecture lingers on Smith's example of mathematicians. Mathematicians are usually obscure. The public does not understand what they do. Yet they continue doing it. Why? Because the motive is not public applause as such. The motive is to be the kind of person whose work is genuinely worthy. Shiller uses this example to argue that finance, like other mature professions, depends heavily on people who have internalized that desire for deserved approval.

The claim is not naively idealistic. The lecturer says explicitly that not everyone reaches this mature state. That is why he next introduces personality psychology and the problem of character.

1.2.3 Personality Differences and Manipulability

The discussion of the DSM-IV appears in the lecture for a very specific reason: to complicate the Smithian picture without abandoning it. People are not all the same. Some are more manipulative, more deceptive, and less constrained by remorse than others. The lecture's rough prevalence claims should be quoted cautiously, because the transcript is noisy at this point, but the magnitudes are clear enough to preserve:

$$\Pr(\text{APD} \mid \text{male}) \approx 0.03, \quad (1.1)$$

$$\Pr(\text{APD} \mid \text{female}) \approx 0.01, \quad (1.2)$$

$$\Pr(\text{APD} \mid \text{prison inmate}) \approx 0.40. \quad (1.3)$$

The lecture's descriptive list is more important than the exact percentages: lack of remorse, frequent lying, lack of empathy, superficial charm, shallow emotion, and a manipulative relation to others. These observations are not the mathematical spine of the chapter, but they do explain why the lecturer refuses to rest content with general moral optimism. Finance is populated by real mixtures of character.

From there the lecture broadens again. We are all manipulable, not only by extreme personalities but by ordinary commercial tactics. The example of pricing points, such as \$9.99 instead of \$10.00, is intentionally simple. It is not cited as a moral catastrophe. It is cited to show how competition itself rewards psychological leverage. The lecturer's conclusion is subtle: individual acts may look somewhat manipulative, yet what matters in the long run is the total character of the person or institution through time.

With that introduction behind us, the lecture finally turns to the famous formal core.

1.3 Prospect Theory: Setup and Board Architecture

The lecturer says he wants to start with what is probably the most famous element of behavioral economics: prospect theory. It was introduced by Daniel Kahneman and Amos Tversky as a theory of how people make decisions about prospects, that is, gambles, under uncertainty. The lecture also warns us not to overread what follows. There is a huge literature here, and what we are about to see is only a quick sketch.

Prospect. A prospect is a gamble, or more generally a choice problem under uncertainty.

Prospect theory. In the lecture's own decomposition, prospect theory has two parts: a value function, which describes how gains and losses are experienced, and a weighting function, which describes how probabilities are psychologically processed.

The retained board image is important because it preserves the lecture's pacing. On the left the board reads "Value Function" and "Weighting Function?"; on the right it reads "Prospect theory" and "Kahneman & Tversky." Nothing has yet been turned into a polished formula. The lecture first names the two pieces, then says, "I'll draw a picture," and only then begins the mathematical interpretation.

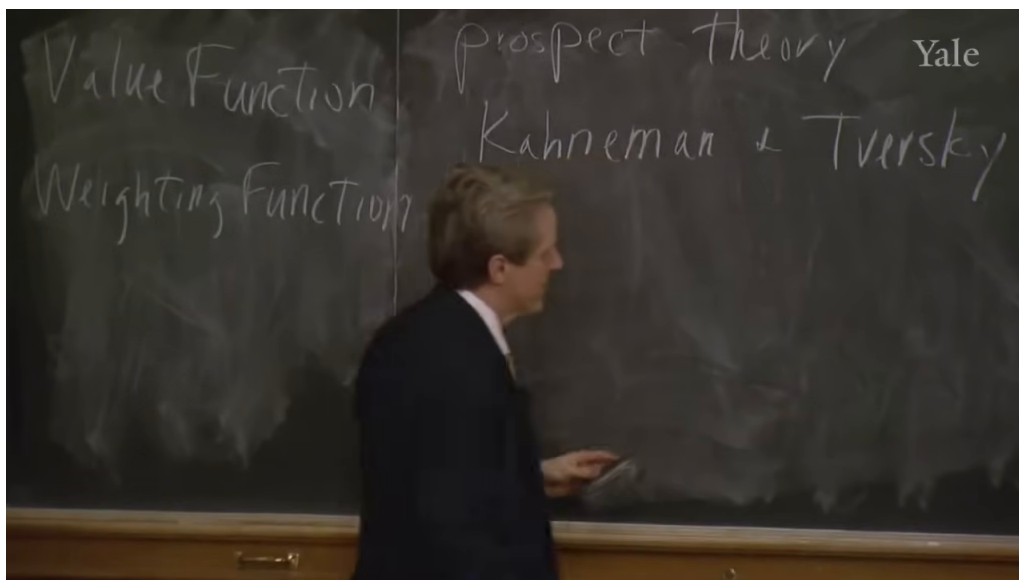


Figure 1.1: Blackboard headings introducing prospect theory, the value function, and the weighting function.

1.4 The Value Function

Now the lecture slows down visibly. The lecturer draws a vertical line and a horizontal line through the middle of the board. The horizontal axis is wealth, or money, or something like that, with zero in the middle. The vertical axis is value, something like utility. The left side is losses; the right side is gains.

For note-writing purposes it is convenient to introduce notation that the board itself does not write explicitly. Let W denote wealth and let r denote the psychological reference point. We then write

$$x = W - r. \quad (1.4)$$

This is a cautious reconstruction for clarity. The board itself labels regions as losses and gains rather than introducing a symbol x .

The lecture's qualitative claims can now be summarized cleanly. The curve passes through the origin, the gain side is concave, the loss side is convex, and there is a kink at the origin:

$$v(0) = 0, \quad x < 0 \Rightarrow \text{losses}, \quad x > 0 \Rightarrow \text{gains}. \quad (1.5)$$

$$v''(x) > 0 \quad \text{for } x < 0, \quad v''(x) < 0 \quad \text{for } x > 0. \quad (1.6)$$

We should be careful about what is and is not being claimed. No closed-form formula is written on the board. What is written, and then verbally interpreted, is a qualitative graph. On the gain side the curve is like diminishing marginal utility in ordinary economic theory. On the loss side it bends the other way. The value of gains and losses is therefore not linear.

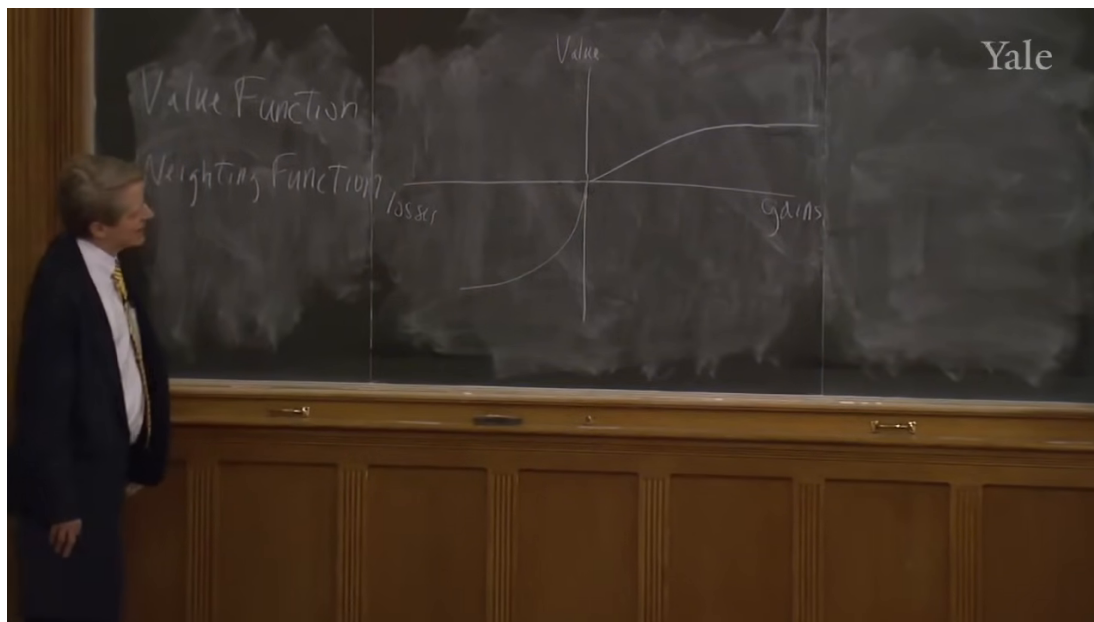


Figure 1.2: Prospect-theory value function with losses on the left, gains on the right, and value on the vertical axis.

1.4.1 Question & Answer

Question. What does the kink at the origin mean, and why does the reference point matter?

Answer. The reference point is the psychological zero from which gains and losses are measured. Usually it is today's wealth, but the lecture insists that it need not be fixed. It can be shifted by framing, that is, by the way a prospect is presented. Once the reference point is set, the kink says that small movements around that point matter disproportionately. We react strongly to little losses near zero. That is why framing changes behavior: the same objective prospect, described relative to a different reference point, can produce a different decision.

A cautious note-writing summary of the lecture's verbal asymmetry is

$$-v(-\varepsilon) > v(\varepsilon) \quad \text{for small } \varepsilon > 0. \quad (1.7)$$

This inequality is not written on the board, but it captures the lecturer's explicit claim that we are "spooked" by small losses and "less encouraged" by small gains.

The lecture's own example is a loss of \$5 on the way to class. Relative to lifetime wealth, \$5 is negligible. Relative to the reference point of "my money this morning," it is vivid and painful. The psychological measurement is local, not actuarial.

From the graph to an insurance market. The lecture immediately turns the graph into an institutional argument. The logic runs as follows.

1. Place the origin at the current reference point, usually today's wealth.
2. Read the kink as heightened sensitivity to small losses near that point.

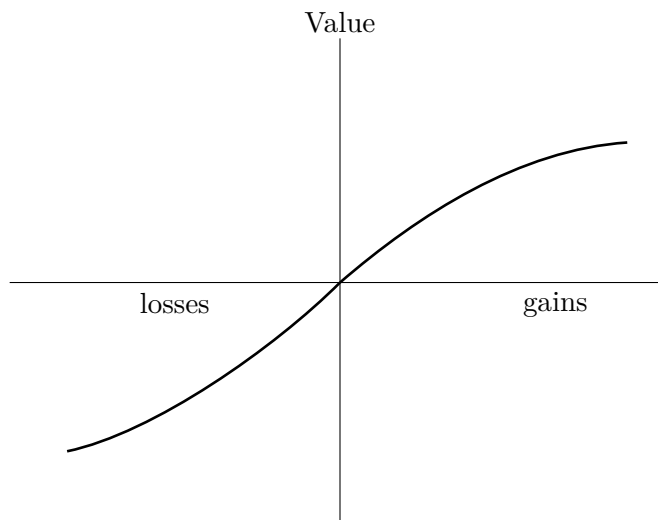


Figure 1.3: Clean qualitative redraw of the prospect-theory value function based on the board sketch.

3. Notice that firms can identify small, salient, bounded losses that consumers are actively imagining.
4. Sell insurance on those narrow losses rather than on the household's largest lifetime risks.
5. The lecture's main examples are funeral insurance, airline flight insurance, and diamond-ring insurance.

This is one of the lecture's sharpest claims. The insurance people *ought* to buy is concentrated on truly large losses, such as the death of a wage earner. But the insurance that often sells most easily is insurance attached to a narrow, emotionally salient event. Funeral insurance has been sold since ancient Rome precisely because it isolates a visible expense that people can picture. Diamond-ring insurance works similarly. It is not the household's central financial vulnerability, but it is vivid.

Framing belongs here as well. Kahneman and Tversky's point, as Shiller states it, is that the same prospect can be presented in different language so as to suggest a different reference point. The prospect itself has not changed. The behavior does.

The lecturer also adds an important qualification before moving on. These distortions are real, but not total. People eventually learn. They come to distrust crude manipulation. So even here, at the most formal point in the lecture, the narrative is still double-sided: the bias is real, and yet the institution adapts around it.

1.5 Weighting Probabilities and Rare Events

The second half of prospect theory concerns not value but probability. The lecture's point is that objective probabilities and psychological decision weights are not the same thing. A probability is a number between 0 and 1, but human beings do not naturally think in a smooth continuum of such numbers.

For note-writing purposes we introduce a weighting function $w(p)$, again with the warning that the

lecture gives only a qualitative sketch:

$$p \in [0, 1], \quad w(0) = 0, \quad w(1) = 1. \quad (1.8)$$

$$w(p) \approx 0 \quad \text{for very small } p, \quad w(p) \approx 1 \quad \text{for very large } p. \quad (1.9)$$

The crucial part is the middle. Very small probabilities may be rounded down to zero. Very large probabilities may be rounded up to one. But if a small probability is not rounded away, it may be exaggerated sharply. The lecture describes the simplest version of the weighting function as almost a broken-line psychological sketch, not as a smooth calibrated formula. That matters. We should not import a polished textbook parameterization where the lecture gives only a rough intuition.

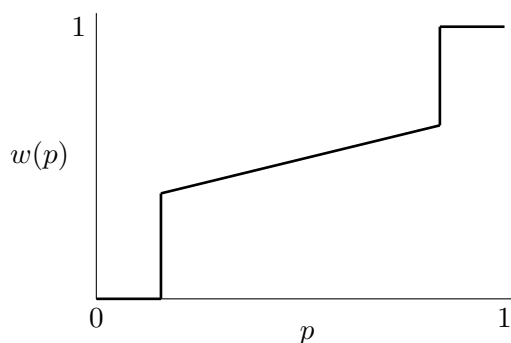


Figure 1.4: Transcript-based schematic of the lecture’s qualitative weighting function. The retained screenshots do not preserve this graph, so the diagram is an interpretive reconstruction rather than frame-backed evidence.

The lecture’s analogy is memorable: our minds handle probabilities as if there were only a few categories. Roughly speaking, the psychological categories are “cannot happen,” “maybe,” and “will happen.” That is why tiny probabilities can be either ignored or inflated.

The institutional example is airline flight insurance. The crash probability of a commercial flight is extraordinarily small. Most travelers mentally round it to zero and move on. But some do not. For them the event takes on a disproportionately large psychological weight, especially at the moment of boarding. That is why airport vending machines once sold flight insurance at highly unfavorable prices: the machines did not need every traveler, only the subset who could not bring themselves to round the probability down.

The lecture immediately complicates its own example. Those machines have largely disappeared. Why? Because the error is not immutable. People learn, experience accumulates, and trusted intermediaries matter. If my ordinary insurance agent does not recommend a special airport policy, I begin to ignore the vending machine. That is why the lecture briefly invokes Gerhard Gigerenzer and then Nick Barberis’s experiments with highly quantitative students: the tendencies are real, but they are not destiny.

The lecturer’s own transition is telling: prospect theory explains a great deal, but it does not explain everything. At that point he explicitly says, in effect, let me move on.

1.6 Other Behavioral Mechanisms

The rest of the lecture accelerates. It does not try to build one grand system out of the remaining material. Instead it moves through a sequence of mechanisms, each connected to some financial surface.

1.6.1 Regret and Gambling

Regret theory is introduced first. It is related to prospect theory, but it isolates something more specific: the painful experience of realizing that one has made a mistake. The lecture's phrase is the ordinary one, "I was kicking myself." Regret is therefore not just a bad outcome. It is a bad outcome with ego attached to it. That can distort decision-making, because we may avoid actions whose later failure would feel humiliating.

The next topic is gambling. The lecture treats gambling as a human universal rather than as a narrow pathology. Anthropologists find it in every society, and the lecture quotes rough prevalence figures from one U.S. study:

$$\Pr(\text{gambled at least once in the year}) \approx 0.61, \quad (1.10)$$

$$\Pr(\text{compulsive gambler} \mid \text{male}) \approx 0.011, \quad (1.11)$$

$$\Pr(\text{compulsive gambler} \mid \text{female}) \approx 0.005. \quad (1.12)$$

Again the numbers are supportive rather than central. The real point is that gambling is psychologically connected to competence, self-image, stimulation, and even relief from depression. In that sense it can resemble an addiction. Yet Shiller does not reduce the stock market to pathology. The same appetite for stimulation can be channeled into productive enterprise. That is one of the lecture's recurring moves: a dangerous psychological impulse can sometimes be organized into something socially useful.

1.6.2 Overconfidence: a Worked Example

The lecturer then slows down again for a classroom experiment. Overconfidence is not presented as mere swagger. It is operationalized through the notion of a 90% confidence interval. If a subject reports an interval $[L, U]$ for an unknown quantity X , then the intended meaning is

$$\mathbb{P}(L \leq X \leq U) = 0.9. \quad (1.13)$$

In ordinary language, if we keep giving honest 90% confidence intervals, we should be right nine times out of ten.

The lecture asks for three such intervals: one for world population, one for the mass of the Earth, and one for the number of languages in the world. The stable numerical answers preserved in the transcript are

$$X_{\text{world population}} = 6,901,330,581, \quad (1.14)$$

$$M_{\text{Earth}} = 5.974 \times 10^{24} \text{ kg}, \quad (1.15)$$

$$N_{\text{languages}} = 6909. \quad (1.16)$$

For the Earth-mass question the lecture briefly changes units. The stable conversion is

$$1 \text{ metric ton} = 10^3 \text{ kg.} \quad (1.17)$$

The transcript around the verbal conversion is noisy, so the kilogram figure is the safer quantity to preserve.

Now comes the diagnostic step. The class raises hands if the true value falls inside the interval. The reported frequencies are only rough show-of-hands counts, but they are sufficient:

$$\text{hit rate}_{\text{world population}} \approx 0.8, \quad (1.18)$$

$$\text{hit rate}_{\text{Earth mass}} \approx 0.1, \quad (1.19)$$

$$\text{hit rate}_{\text{languages}} \approx 0.1. \quad (1.20)$$

The inference is straightforward and worth stating explicitly.

1. A nominal 90% interval should contain the truth about 90% of the time.
2. The realized inclusion rates are far below 0.9.
3. Therefore the intervals were too narrow.
4. Therefore subjects knew less than they thought they knew.

That is overconfidence in an operational sense. The lecture's explanation is also important. We tend to think from the first perspective that comes to mind. We do not canvass enough ways in which the world might surprise us. Since imagination is narrower than reality, the interval is too narrow as well.

Shiller then widens the application. Overconfidence does not stop with the self. It extends to our friends, to leaders, to central bankers, and to charismatic CEOs. This is where the lecture invokes Rakesh Khurana's *Search for the Charismatic CEO* and Taleb's *Fooled by Randomness*. Chance success is quickly rewritten as genius, and institutions can then be reorganized around that illusion.

1.6.3 Cognitive Dissonance

Cognitive dissonance is presented as a bias in belief maintenance. It is painful to admit that a belief to which we are attached is wrong, so we selectively seek confirming evidence and evade disconfirming evidence. The lecture's early example is the car experiment: after buying a car, people attend to advertisements for the car they chose and avoid advertisements for the alternatives they rejected.

The same mechanism appears in finance. Shiller cites work by Will Goetzmann and coauthors on mutual-fund investors. When a fund performs badly, many investors do sell. But some hang on, and the striking point is that they often do not even have an accurate impression of how badly the fund has done. Unpleasant evidence is not neutrally processed. It is blocked out.

The lecture then turns to a more institutionally awkward example, drawn from Mullainathan's work on financial advice. Actors presented advisers with extremely concentrated portfolios: all money in money-market funds, or all money in tech stocks, or all wealth in one's employer's stock. Professionally, an adviser should challenge the concentration. In practice many advisers did not, especially when the client's position seemed to express a deep personal conviction.

1.6.4 Question & Answer

Question. If advisers know that clients are biased, when should they challenge those beliefs and when should they work around them?

Answer. The lecture does not offer a formula, and that is precisely the point. If an adviser immediately attacks a new client's deepest convictions, the client may simply leave. The adviser may therefore soften the challenge, delay it, or fail to make it at all. That may look like cowardice, but the lecture treats it as a real professional dilemma. A bluntly correct intervention can be institutionally ineffective. The moral problem is therefore not only what is true, but how truth can be introduced without losing the relationship through which correction might eventually occur.

1.6.5 Anchoring, Representativeness, and Social Contagion

Anchoring comes next. The lecture retells Kahneman and Tversky's wheel-of-fortune experiment. Subjects are first exposed to a random number and then asked a numerical question. The random number has no logical bearing on the answer, yet answers drift toward it. When ambiguity is genuine, judgment is pulled by whatever salient number happens to be nearby.

Representativeness is a different distortion. Here the mind overreads remembered patterns as if they were reliable guides. The lecture's example is the head-and-shoulders pattern in technical analysis. A rare pattern, once noticed, is then seen too often. This creates room for manipulation. If traders believe a head-and-shoulders formation predicts decline, a manipulator in a thin market can engineer price trades that mimic the pattern and then short into the resulting sell signal.

The lecture's final behavioral mechanism is social contagion. This is really social psychology rather than individual psychology. Our beliefs are interdependent. Herd behavior does not usually feel like conscious imitation. It feels like ordinary common sense inside a shared field of opinion. Shiller invokes Durkheim's phrase "collective consciousness" to capture this. The practical conclusion is plain enough: shared narratives and shared memory can generate large market swings.

1.7 Institutions, Shared Value, and Integrity

The lecture ends where it began, but with more detail behind the optimism. There is manipulation. There is exploitation. There are antisocial personalities. There are recurrent errors in value perception, probability weighting, confidence, updating, and social judgment. But these are not the only facts.

Institutions evolve. Some manipulations become crimes. The lecture's own example is the deliberate manufacture of a head-and-shoulders pattern in order to profit from the public's response. That is not merely clever. It is market manipulation, and the lecturer points out that it is prosecuted.

Beyond formal law, there is still the older Smithian theme. People who rise to important positions in finance often do so partly because they have an internal compass. The lecturer explicitly returns to praiseworthiness here. The desire to be the kind of person one ought to praise is not a decorative moral. It is part of the explanation of why financial markets do not become wholly dominated by exploitation.

The lecture's last two references reinforce this. Michael Porter and Mark Kramer are cited for the idea of shared value: mature business leadership does not think only in terms of narrow short-run shareholder maximization. It understands that long-run value is created with society, not against it. Anna Bernasek's *The Economics of Integrity* makes the same point through a different example. We are safe drinking milk not only because of regulation, but because the people producing and handling it usually think it is obvious that one should keep it clean. Integrity is not an afterthought. It is one of the hidden supports of economic life.

That is the lecture's final balance. Behavioral finance matters because people are vulnerable to error. It also matters because finance is an institutional achievement that learns how to live with that vulnerability.

1.8 Summary

We began where the lecture begins: with the claim that finance must be built for real people rather than idealized agents. From there the lecture deliberately refused a purely cynical interpretation. Human weakness creates opportunities for exploitation, but reputation, morality, regulation, and character all limit how far exploitation can organize the system.

Only then did the lecture turn to prospect theory. The value function gives the chapter its first mathematical spine: gains and losses are measured relative to a reference point, the gain side is concave, the loss side is convex, and the origin is kinked. The weighting function gives the second: objective probabilities are psychologically rounded, simplified, and sometimes exaggerated. Around that core the lecture assembles regret, gambling, overconfidence, cognitive dissonance, anchoring, representativeness, and social contagion.

The concluding claim is not that markets are clean, and not that they are rational in any simple sense. It is that financial institutions evolve, people learn, regulation matters, and integrity remains economically important. Behavioral finance therefore enlarges finance theory. It does not replace the institutional world with a list of biases; it explains why that world must be designed for the complicated creatures who live inside it.